

Adapting SAM3 for 3D fruit counting with cross-view contrastive learning and Hough voting[☆]

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ARTICLE INFO

Dataset link: <https://github.com/kkk0216/SAM3-3DFruit>

Keywords:

3D Gaussian

Fruit counting

Vision foundation model

Occlusion

Agricultural robotics

ABSTRACT

Fruit detection and 3D localization have been shown to be critical for precision agriculture applications, such as yield estimation and robot harvesting tasks. However, real-world agricultural settings can be associated with high density of fruits, strong occlusions, and variable illumination, leading to difficulty in accurately detecting and localizing fruits. Although there exist 2D approaches dedicated for fruit counting, these approaches may easily merge adjacent fruits due to projection effects in dense occlusions, thus limiting their effectiveness. Motivated by this, in this paper, we explore the problem of 3D fruit detection and localization with vision foundation model adaptation and 3D Gaussian-based scene representation. Particularly, we use low-rank adaptation to leverage SAM3 to generate high-quality instance masks with different crops and illuminations. Meanwhile, to address noisy pseudo label issue, we adopt cross-view contrastive learning with uncertainty weighting for enhanced 3D feature representation. Moreover, we devise a Hough voting procedure in the 3D Gaussian space to integrate multi-view evidence of occluded fruits, followed by clustering to estimate the number of fruits as well as their 3D locations. The proposed approach has been extensively tested with thousands of fruits from synthetic orchard scenes, real apple orchard datasets, as well as greenhouse sweet pepper scenes. Experiment results show significant performance gains for densely occluded cases, with the F1-scores reaching 0.971 and 0.977 for plum and mango, respectively. Meanwhile, our approach can achieve more than a twofold boost for 3D instance segmentation over FruitNeRF. More notably, the adapted weights and the post-processing settings transfer from the open-field apple orchard to greenhouse sweet pepper scenes, where only the SAM3 text prompt is changed and the 3D Gaussian representation is still optimized for each new scene.

1. Introduction

Modern precision agriculture heavily utilizes automated systems to facilitate the observation and analysis of crop traits across different field environments (Tian et al., 2020). Fruit counting is crucial, directly impacting yield estimation, harvest planning, and robotic operations (Tu et al., 2025). In actual production, manual counting is not only time-consuming but also often unreliable in large-scale orchard scenarios. Therefore, vision-based methods have been widely adopted, primarily due to the rapid development of deep learning in object detection tasks.

With the recent progress of deep learning and vision foundation models (VFM), agricultural perception tasks have become increasingly

automated (Devanna et al., 2025; Li et al., 2026). Trained on large-scale datasets, these models exhibit strong general visual representation capabilities and provide effective support for object detection, segmentation, and recognition in complex agricultural scenes. However, agricultural images differ significantly from general natural images. In orchards and greenhouses, dense foliage, varying illumination, occlusion, and similar appearances of fruit often reduce the effectiveness of general-purpose models when applied directly. Therefore, this study focuses on adapting vision foundation models to agricultural scenes and integrating them with three-dimensional spatial reasoning for more reliable fruit perception. The overall pipeline of the proposed method is shown in Fig. 1.

[☆] This article is part of a Special issue entitled: 'Foundation Models in Agriculture' published in Computers and Electronics in Agriculture.

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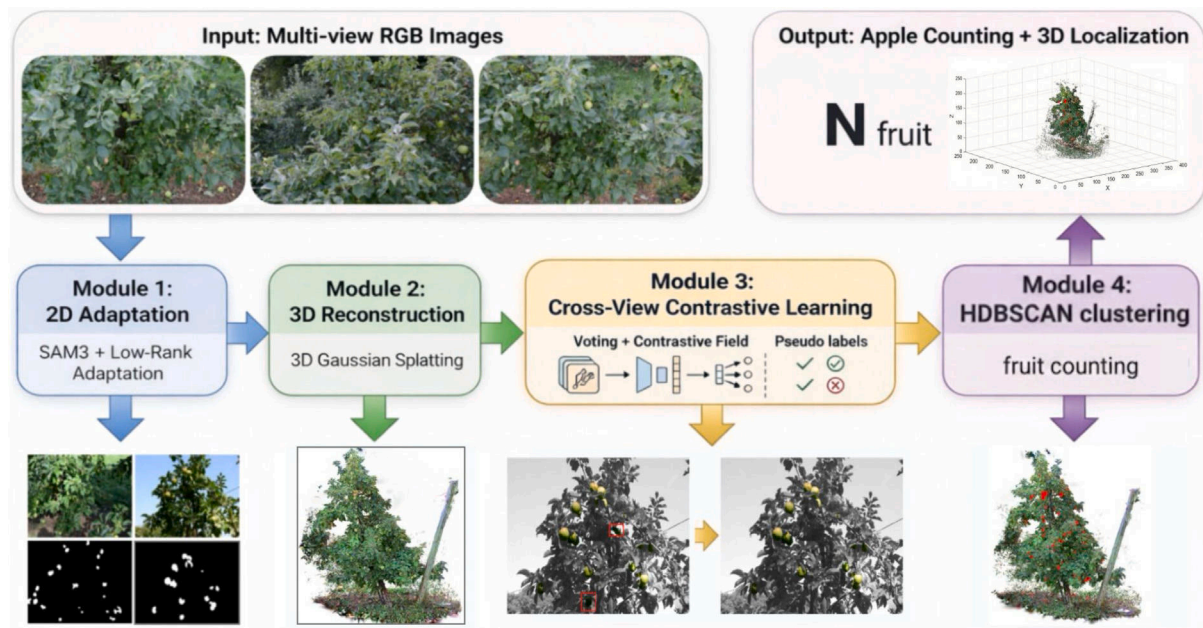


Fig. 1. Pipeline overview. SAM3 is adapted to produce 2D instance masks from input images, which guide 3D Gaussian Splatting reconstruction. Cross-View Contrastive Learning and a voting mechanism refine feature consistency and estimate fruit centers, with final counts and 3D positions obtained via clustering.

Early research primarily relied on handcrafted features or depth detectors to achieve 2D image detection from a single RGB image (Sapkota et al., 2024; Kang and Chen, 2020). These methods performed well under controlled conditions, but their performance significantly degraded in real orchard or greenhouse environments. In real-world scenes, fruits are typically densely distributed and often occlude one another, while short-term illumination variations can be drastic, with adjacent fruits sometimes separated by only a few centimeters or even overlapping (Gené-Mola et al., 2020). Representative examples include plum scenes in FruitNeRF (Meyer et al., 2024) and apple scenes in Fuji-SfM (Gené-Mola et al., 2020).

Recent studies on fruit counting and yield estimation based on the YOLO family of detectors have achieved substantial progress in tasks such as young apple fruit detection (Z. Liu et al., 2024), grape picking-point localization (Wang et al., 2025), passion fruit yield estimation (Tu et al., 2025), cotton boll number prediction (Tan et al., 2025), and lightweight detection networks for orchard deployment (Song et al., 2025; Zhang et al., 2025). However, most existing approaches still rely on single-view detection or video-tracking paradigms, which exhibit limited robustness under severe occlusion among adjacent fruits (Liu et al., 2025). As highlighted by Z. Li et al. (2025) in a recent review of fruit 3D reconstruction, the transition from 2D detection to 3D perception is becoming an important trend in precision agriculture for addressing densely occluded fruit scenes.

Multi-view geometry offers a potential solution to this problem by improving reliability through the fusion of observations from different perspectives. Structure from Motion (SfM) can reconstruct camera poses and sparse scene geometry, thereby enabling the alignment of detection results across multiple views (Gené-Mola et al., 2020). Neural scene representation methods further extend this approach. FruitNeRF (Meyer et al., 2024), for example, combines segmentation with volumetric rendering to reduce duplicate counting to some extent. However, its high computational cost and lengthy optimization process limit its applicability in practical agricultural workflows (Z. Li et al., 2025).

In recent years, 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) has gained increasing attention as a more efficient scene representation. It uses explicit Gaussian primitives to describe the scene, each carrying

information on position, shape, opacity, and color. Compared to the implicit model NeRF, this representation renders faster and facilitates geometric inference. In agricultural applications, these Gaussian primitives can be filtered or grouped to identify individual fruits without relying on expensive volume queries (Jiang et al., 2025). Existing research has applied this technology to plant phenotypic analysis, such as cotton boll localization and strawberry reconstruction (Jiang et al., 2025; J. Li et al., 2025). Regarding vision foundation models, the Segment Anything Model (SAM) (Kirillov et al., 2023) has demonstrated stable zero-shot segmentation capability, while its successor SAM3 (Carion et al., 2025) further supports text-driven open-vocabulary instance segmentation, allowing segmentation tasks to be guided by natural language descriptions. Early studies adapting vision foundation models to agricultural applications have shown promising potential. For example, Devanna et al. (2025) combined SAM with Grounding DINO for zero-shot annotation of grape bunches; Tan et al. (2025) integrated SAM with a density-map model for cotton boll counting; and Li et al. (2026) applied SAM to agricultural field parcel extraction through adaptive mask refinement. However, directly applying these general-purpose foundation models to dense fruit scenes remains challenging. The visual distribution of orchards and greenhouses differs substantially from that of the natural images used for pre-training: fruits frequently occlude each other, shadows and illumination variations blur object boundaries, and dense spatial distributions make instance separation more difficult. These factors collectively introduce domain bias and reduce segmentation accuracy.

Training such models from scratch is computationally expensive and requires large-scale annotations, making it impractical for most agricultural scenarios. In contrast, Low-Rank Adaptation (LoRA) (Hu et al., 2022) provides a parameter-efficient fine-tuning strategy that updates only a small number of parameters while effectively mitigating domain bias. In this work, we adapt SAM3 using LoRA to improve its segmentation performance in agricultural scenes while preserving its general visual knowledge. Nevertheless, improving 2D segmentation alone is insufficient, because the resulting masks are further used for 3D reconstruction. If the masks contain noise or if instance labels are inconsistent across different viewpoints, these errors can propagate into the 3D representation and ultimately affect fruit counting. To address this issue, we tightly integrate cross-view 3D reasoning with the vision

foundation model. Specifically, we use 3D Gaussian representations to aggregate multi-view information, and design an uncertainty-weighted cross-view contrastive learning strategy to reduce the influence of noisy pseudo-labels. Meanwhile, a voting mechanism is introduced to fuse cross-view spatial cues, enabling stable fruit center estimation even under occlusion.

The main contributions of this paper are summarized as follows. First, we propose SAM3-Adapt, a LoRA-based parameter-efficient adaptation method for improving fruit instance segmentation with SAM3 in agricultural scenes. It keeps the general representation ability of the vision foundation model while reducing the domain gap caused by agricultural images. Second, we introduce an uncertainty-aware cross-view contrastive learning module, which reduces the effect of noisy pseudo-labels on 3D feature learning and makes fruit instance features more consistent across views. Third, for fruit counting and 3D localization under occlusion, we add a Hough voting mechanism to the 3D Gaussian representation. By combining spatial evidence from multiple views, it helps estimate fruit centers and separate instances more reliably. In addition, we build a fruit perception dataset covering different agricultural scenes and crop types to evaluate the method under different environments. Finally, by applying the same adapted weights and post-processing settings from open-field orchards to greenhouse sweet pepper scenes, with only the text prompt changed, we demonstrate its transferability for cross-crop 3D fruit perception.

2. Materials and methods

2.1. Dataset construction and preprocessing

We constructed datasets for three types of agricultural scenarios: synthetic scenes, open-air apple orchards, and greenhouses. Experiments were conducted using 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) and the vision foundation model SAM3 (Carion et al., 2025) to examine the model's performance under different lighting conditions, occlusion levels, and fruit densities, while also verifying the transferability of the proposed method across different visual environments.

2.1.1. Multi-source dataset integration

Although computer vision in agriculture has progressed rapidly in recent years (Tian et al., 2020), high-quality 3D agricultural datasets remain scarce. Existing works mainly address 2D detection (Sapkota et al., 2024; Z. Liu et al., 2024) or segmentation (Kang and Chen, 2020), and therefore provide limited geometric information (Z. Li et al., 2025). Collection of 3D agricultural datasets is expensive, and it requires a multi-view collection procedure, precise calibration, and a stable reconstruction algorithm. In addition, different datasets may use different sensors, making comparisons between datasets challenging. In this work, we combine three complementary public agricultural datasets, each of which represents various agricultural scenes and difficulties. Basic statistical properties of the used datasets are listed in Table 1.

The FruitNeRF (Meyer et al., 2024) dataset contains images of synthetic and real-world scenes. Its synthetic part was created using Blender, and includes six classes of fruits: apple, plum, lemon, pear, peach, and mango, providing 2942 annotated fruit instances, along with exact 3D location annotations. Images of 583 real-world fruit instances taken using the Nikon D7100 camera at a resolution of 6000×4000 were also included in this dataset. The dataset provides relatively simple scenes with clearly defined geometric structure and stable camera motion which makes it good for evaluating performance of the considered tasks under controlled conditions. However, it has low diversity of scenes.

The Fuji-SfM (Gené-Mola et al., 2020) dataset consists of images taken from a commercial Fuji apple orchard. In contrast to the FruitNeRF dataset, this one includes a much higher diversity of outdoor scenes that have varying lighting conditions, dense foliage and occlusions. It includes images of one outdoor scene taken using a Sony

Alpha 6000 camera at a resolution of 4000×3000 which contain 1455 annotated fruit instances. Due to the absence of annotations on exact 3D fruit locations, the dataset becomes challenging for solving tasks of fruit instance segmentation, localization, and 3D scene understanding.

The BUP20 (Pan et al., 2023) dataset was acquired using a mobile robot with an attached Intel RealSense D435i RGB-D sensor, capturing RGB-D images and odometry information in a greenhouse environment with sweet peppers. This dataset differs from Fuji-SfM in the presence of depth cues and an indoor structured environment, making it a good choice for multimodal perception evaluation. However, it lacks annotated fruit instances and 3D fruit location labels.

Overall, these three datasets complement each other well. FruitNeRF provides accurate geometric annotations on controlled scenes, while Fuji-SfM brings challenging outdoor orchard conditions. In turn, BUP20 dataset adds RGB-D information from a greenhouse environment to our benchmark.

2.1.2. Data standardization and instance annotation

The datasets used in this article are sourced from different sources, and the formats of the sensor and dataset vary. To ensure data consistency, we first converted all camera parameters to a COLMAP-compatible format. For data sets that contain depth, for example BUP20, the 3D points were created using back-projection of the depth map. However, for RGB data only, as in Fuji-SfM, the cameras were first estimated and point clouds were then built using COLMAP.

With respect to the generation of ground truth (GT) for each dataset, we took various strategies depending on the annotation conditions of each dataset. In the case of FruitNeRF's synthetic subset, we directly utilized the official annotation information in Blender, which includes fruit instance's identity and 3D coordinate. Hence, these annotations do not depend on our method and are not generated from SAM3 results.

For real-world datasets where no complete 3D annotations can be found, we followed a semi-automatic annotation procedure as depicted in Fig. 2. Firstly, we employed SAM3 to get rough candidate instance segmentation results, and then manually corrected each image by using VGG Image Annotator (Dutta and Zisserman, 2019). The process includes excluding false positives, adding missing fruits, splitting stuck instances in dense regions, and correcting inaccurate instance masks. We would like to note that the predicted results from SAM3 were used only for initialization purpose but not directly taken as ground truth.

In addition, we manually refined masks in heavily occluded and dense regions by using the combination of multiview information and reconstructed 3D structure, as illustrated in Fig. 2. Specifically, we keep a fruit when it is occluded in one view but is visible in other views and is consistent with the 3D reconstruction result. And, if an instance cannot be distinguished at all, then it will be removed from GT for 3D localization evaluation.

2.2. SAM3 in agricultural scenes

2.2.1. Limitations of open-set detection for 3D fruit reconstruction

Closed-set detectors are unsuitable for 3D fruit reconstruction, because they require every target category to be fixed before training and degrade sharply once the deployment distribution shifts. Detectors of the YOLO family (Wang et al., 2024) perform stably under a known data distribution, but their generalization ability decreases markedly in unfamiliar environments or for new species unless data are re-collected and the model is fine-tuned. Open-vocabulary detection alleviates this problem by locating objects of unseen categories from textual cues. Grounding DINO (S. Liu et al., 2024) is a representative method: it aligns visual and linguistic features so that detection no longer depends on a predefined category list, which gives it strong semantic generalization ability. However, directly applying such open-vocabulary detection methods to 3D reconstruction still has significant limitations. A key issue is the relatively coarse-grained output format, primarily providing

Table 1

Statistics of the four datasets. GT Fruits: number of annotated fruit instances; GT 3D Pose: availability of 3D coordinates; “-”: incomplete annotation.

Dataset	Source	Fruit type	Scenes	GT Fruits	GT 3D Pose
FruitNeRF (Syn)	Synthetic (Blender)	6 types	6	2942	✓
FruitNeRF (Real)	Real (Nikon D7100)	Apple	3	583	×
Fuji-SfM	Real (Sony α 6000)	Apple	1	1455	×
BUP20	Real (RealSense D435i)	Sweet pepper	1	-	×

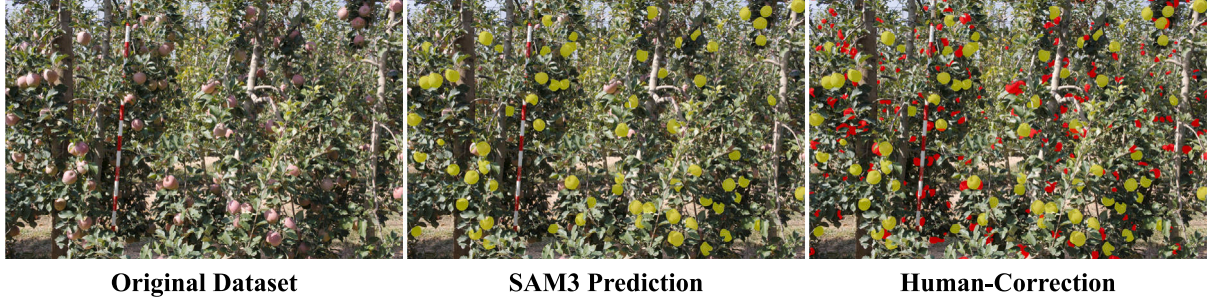


Fig. 2. The annotation refinement pipeline. The annotation refinement pipeline begins with a raw image; the process proceeds through SAM3 prediction, ending with manual correction.

supervision information in the form of bounding boxes. In real agricultural environments, fruits are typically densely distributed and heavily occluded, leading to overlapping bounding boxes between adjacent targets. This results in uncertainty in label assignment and makes it difficult to accurately depict the true geometric boundaries of the fruits. More importantly, in agricultural scenarios, these methods are prone to missing detections, false detections, or category confusion. These errors already exist in the 2D stage and are further amplified after multi-view fusion. For 3DGS segmentation, high-quality segmentation relies on accurate pixel-level supervision signals with cross-view consistency. However, the bounding boxes provided by open-vocabulary detectors are not only coarse-grained but also contain detection errors, making it difficult to maintain consistency across multiple views. These combined problems limit the accuracy of geometric detail recovery, ultimately affecting the accuracy of fruit counting and localization.

2.2.2. Adapting SAM3 to agricultural scenes

Text-prompted segmentation alone is insufficient for agricultural imagery, because foundation models pre-trained on generic natural images underperform task-specific supervised models when the target domain has distinctive visual statistics (Kirillov et al., 2023; Li et al., 2023). SAM3 (Carion et al., 2025) employs a text-based segmentation mechanism, in which a natural-language description drives pixel-level localization and segmentation of all matching instances in an image. Compared with interactive paradigms that rely on manual clicks or bounding boxes, this mechanism generalizes better in open-vocabulary settings. Its pre-training data, however, consist mainly of generic natural images, so zero-shot segmentation in a specialized domain often falls short of expectations.

The visual characteristics of agricultural scenes further amplify this problem. In orchards and greenhouses, fruits of the same type have highly similar colors and textures, resulting in severe mutual occlusion, making it difficult to distinguish instance boundaries in dense areas. Under these conditions, SAM3's text-based prompt mechanism struggles to effectively separate neighboring fruits, easily leading to mask adhesion or boundary drift, and failing to provide stable and consistent pixel-level supervision for subsequent multi-view 3DGS segmentation. In addition, the appearance of fruits varies greatly within a class due to differences in variety, growth stage and light conditions. General text descriptions are difficult to fully cover these fine-grained out-of-domain samples, which naturally affects the robustness of segmentation (Gurav et al., 2023).

2.2.3. Adaptive vision foundation model based on low-rank fine-tuning

Parameter-efficient fine-tuning offers a practical route to close this domain gap, because it adapts a foundation model to a new visual domain at a small fraction of the cost of full fine-tuning while preserving the pre-trained representation (Hu et al., 2022). We therefore employ Low-Rank Adaptation (LoRA) (Hu et al., 2022) to fine-tune the vision foundation model SAM3 (Carion et al., 2025). LoRA does not update all parameters of the model; it inserts small learnable matrices and keeps the pre-trained weights fixed, so the model can absorb agricultural data without losing its general visual capability. For a weight matrix $W_0 \in \mathbb{R}^{d \times k}$ in the SAM3 image encoder, LoRA introduces two low-rank matrices $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$, where $r \ll \min(d, k)$. The forward computation can be written as:

$$h = W_0 x + B A x \quad (1)$$

Indeed, these update terms are injected in the linear projections of Q, K, and V for the Transformer blocks. As shown in Table 2, an ablation study was conducted regarding the LoRA rank r . We observe that the rank $r = 8$ outperforms all other choices, while introducing just 2.95M trainable parameters, representing 0.1% of all parameters in SAM3. The mask decoder, on the contrary, remains unchanged for the segmentation task. The complete procedure is depicted in Fig. 3.

The LoRA rank ablation demonstrates that parameter-efficient adaptation can substantially improve segmentation performance with only a small number of trainable parameters. Even at a low rank of $r = 2$, the adapted model achieves a clear improvement over the Grounded-SAM2 baseline, increasing the average mIoU from 0.561 to 0.737, as shown in Table 2. Among different rank settings, $r = 8$ obtains the best average mIoU of 0.856 with only 2.95M trainable parameters, indicating the most effective trade-off between segmentation accuracy and computational cost. Increasing the rank from $r = 8$ to $r = 16$ slightly improves the performance on the lemon category, but brings negligible improvement in the overall average mIoU. When the rank is further increased to $r = 32$, the average mIoU decreases to 0.846, while the training time increases by approximately 76%. Therefore, we adopt $r = 8$ as the default LoRA rank in our experiments.

Training is divided into two phases. In the first phase, all pre-training parameters are fixed, and only the LoRA module is updated. This allows the model to learn domain-specific content, such as dense regions and different lighting conditions. In the second phase, the last and penultimate layers of the image encoder are unfrozen and fine-tuned together. This allows learning higher-level information without altering the lower-level content.

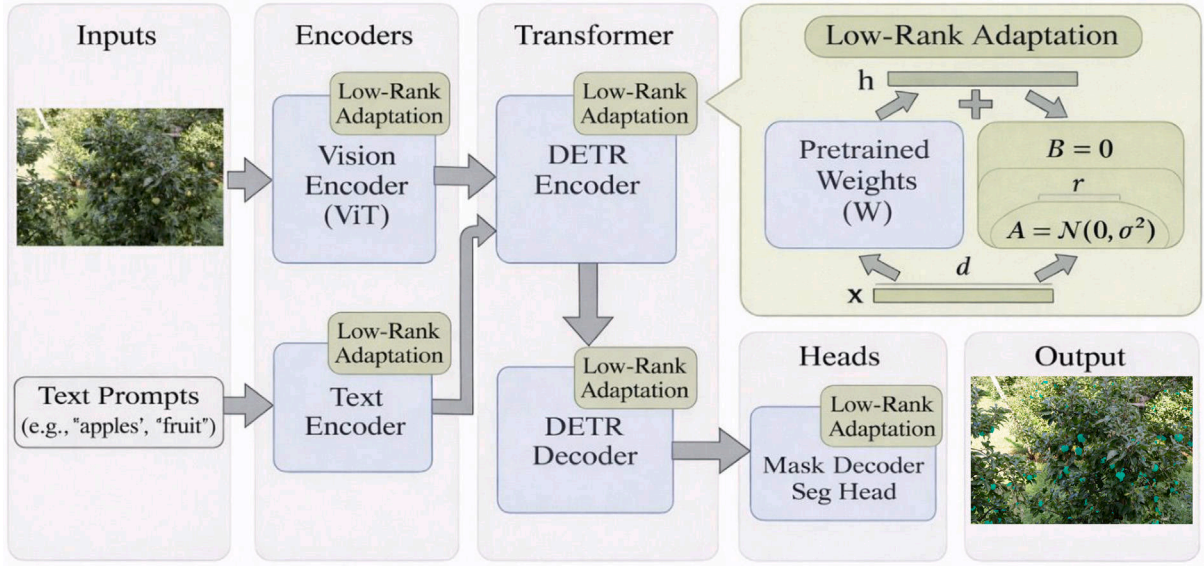


Fig. 3. Overview of the SAM3 adaptation process. Insert low-rank modules into the image encoder, keep the remaining pre-trained layers fixed, and only update the selected layers during training.

Table 2

LoRA rank ablation for SAM3 adaptation, evaluated by 2D segmentation mIoU on the six FruitNeRF synthetic categories. Best value per column in bold.

Method	Params (M)	Apple	Plum	Lemon	Pear	Peach	Mango	Avg. mIoU ↑	Train (h)
Grounded-SAM2	–	0.659	0.382	0.556	0.519	0.662	0.588	0.561	–
SAM3 (zero-shot)	0	0.712	0.428	0.601	0.567	0.704	0.624	0.606	–
Ours, $r = 2$	0.74	0.819	0.604	0.748	0.715	0.806	0.732	0.737	2.1
Ours, $r = 4$	1.47	0.871	0.731	0.815	0.788	0.858	0.798	0.810	2.4
Ours, $r = 8^a$	2.95	0.898	0.806	0.859	0.837	0.887	0.847	0.856	2.9
Ours, $r = 16$	5.90	0.901	0.802	0.864	0.834	0.885	0.844	0.855	3.6
Ours, $r = 32$	11.80	0.893	0.787	0.857	0.829	0.880	0.831	0.846	5.1

^a Marks the default configuration.

The segmentation loss combines Focal Loss and Dice Loss to balance class imbalance and boundary quality:

$$\mathcal{L}_{\text{seg}} = \lambda_{\text{focal}} \mathcal{L}_{\text{Focal}} + \lambda_{\text{dice}} \mathcal{L}_{\text{Dice}} \quad (2)$$

To improve robustness, we employ common image data augmentation techniques, such as random cropping, horizontal flipping, color jittering, and Gaussian blur. These simulate variations in agricultural conditions caused by different lighting, angles, or sensor noise. Training is performed using the AdamW optimizer on an NVIDIA A6000 GPU.

2.2.4. Evaluating adaptation of vision foundation models

After LoRA fine-tuning, SAM3 produces more stable and spatially consistent instance masks in both synthetic and real agricultural scenes, especially in densely occluded regions, as shown in Figs. 4 and 5. We compare our method with two segmentation baselines used for agricultural scenes: the original SAM3 and Grounded-SAM2, which pairs Grounding DINO with a SAM-family segmenter (Ren et al., 2024). For a fair comparison, all methods use the corresponding fruit category names as text prompts. The evaluation is carried out on four datasets, including FruitNeRF synthetic scenes, FruitNeRF real-world samples, the Fuji-SfM orchard dataset, and the BUP20 greenhouse dataset.

From the qualitative results, the adapted model gives more complete masks and separates nearby fruits more clearly than the baselines. This is especially noticeable when fruits are clustered together or partly occluded, where instance separation is difficult. By contrast, the original SAM3 and Grounded-SAM2 often produce broken masks or overlapping instance regions, which makes the segmentation less reliable. These results suggest that LoRA fine-tuning helps the foundation segmentation

model better fit agricultural scenes with dense fruit layouts and strong occlusion.

Although LoRA adaptation improves the quality of 2D fruit masks, the generated pseudo-labels still contain errors. In heavily occluded regions, mask boundaries may be inaccurate. Fruits with colors close to leaves, branches, or the background may also be missed or wrongly classified. In addition, because the appearance of the same fruit changes across viewpoints, its pseudo-labels may not always be consistent in different views. These issues make fully reliable 2D supervision difficult to obtain in real agricultural scenes. If such noisy masks are directly used for 3D representation learning, the supervision may become biased and affect feature learning as well as instance separation. To reduce this problem, we use two complementary mechanisms. The contrastive learning module in Section 2.3.1 weakens the influence of unreliable samples and helps learn more discriminative instance-level features. At the same time, the multi-view voting strategy in Section 2.3.3 combines observations from different views to improve spatial consistency. Although these mechanisms cannot remove all segmentation errors, they reduce the effect of noisy 2D masks and provide more stable guidance for 3D fruit localization, centroid estimation, and instance prototype construction.

2.3. 3D Gaussian splatting-based fruit instance segmentation

2.3.1. Cross-view feature learning for Gaussian representations

Refining the 2D masks is necessary but not sufficient, because residual mask noise propagates into the 3D representation and is amplified after multi-view mask fusion. Even after adaptation, the SAM3 masks still show blurred boundaries, missing small fruits, and leaf occlusions.

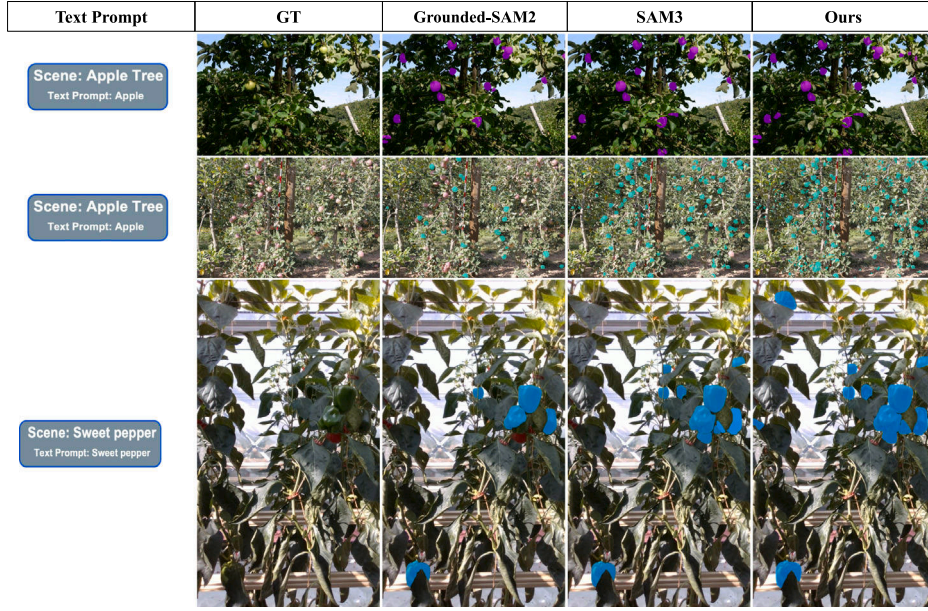


Fig. 4. Segmentation results on real-world scenes, including orchard and greenhouse environments.

To limit the impact of these errors and improve consistency across views, we apply contrastive learning to the Gaussian features, so that Gaussians belonging to the same fruit stay close in feature space while those from different fruits are pushed apart.

3DGS (Kerbl et al., 2023) represents a scene as a collection of N explicit Gaussian primitives $\mathcal{G} = \{g_i\}_{i=1}^N$, where each Gaussian $g_i = \{\mathbf{p}_i, \mathbf{s}_i, \mathbf{q}_i, o_i, \mathbf{c}_i\}$ contains position, scaling, rotation, opacity, and color. Given camera pose, pixel colors are rendered through α -blending:

$$C_u = \sum_{i \in \mathcal{N}} c_i \alpha_i T_i \quad (3)$$

where $\mathcal{N}$ denotes the ordered set of Gaussian primitives whose projections cover pixel u , sorted in ascending depth (front-to-back) along the viewing ray, and the index i enumerates this sorted sequence; $\alpha_i = o_i \exp(-\frac{1}{2} \mathbf{d}^T \Sigma^{-1} \mathbf{d})$ is the Gaussian weight, $T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$ is the cumulative transmittance accumulated over the $i-1$ Gaussians lying in front of the i th one, $\mathbf{d}$ is the vector from the sample point to the Gaussian center, and Σ is the covariance matrix.

We attach a learnable feature vector $\mathbf{f}_i \in \mathbb{R}^{16}$ to each Gaussian, rendered via α -blending:

$$\mathbf{F}_u = \sum_{i \in \mathcal{N}} \mathbf{f}_i \alpha_i T_i \quad (4)$$

We adopt ProtoNCE contrastive learning (Li et al., 2020) to optimize features. Let $\{\Omega_j\}_{j=1}^C$ denote the set of SAM3 instances in the current view, where Ω_j is the set of pixels covered by the j th instance. We compute instance prototypes $\bar{\mathbf{F}}_j = \frac{1}{|\Omega_j|} \sum_{u \in \Omega_j} \mathbf{F}_u$. The contrastive loss is:

$$\mathcal{L}_{\text{contra}} = -\frac{1}{C} \sum_{j=1}^C \sum_{u \in \Omega_j} \log \frac{\exp(\text{sim}(\mathbf{F}_u, \bar{\mathbf{F}}_j) / \phi_j)}{\sum_{l=1}^C \exp(\text{sim}(\mathbf{F}_u, \bar{\mathbf{F}}_l) / \phi_l)} \quad (5)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and ϕ_j is a dynamic temperature parameter. The inner summation $\sum_{u \in \Omega_j}$ runs over all pixels u belonging to the j th instance Ω_j , so that each pixel feature $\mathbf{F}_u$ is contrasted against its own instance prototype $\bar{\mathbf{F}}_j$ (positive) and all other prototypes (negatives). This loss encourages pixel features within the same instance to cluster around their prototype while remaining distant from other prototypes.

The contrastive objective above is constructed independently within each view. The instance partition, the prototypes, and the positive and negative pairs are all defined by the SAM3 masks of the current image,

so no instance correspondence between different views is required. The rendered feature $\mathbf{F}_u$ is obtained by α -weighted accumulation of the per-Gaussian features $\mathbf{f}_i$, and each $\mathbf{f}_i$ is a view-independent, scene-level parameter shared by every view in which the corresponding primitive is visible. The per-view contrastive losses therefore update the same feature vector, and the gradient accumulates over the visible views,

$$\frac{\partial \mathcal{L}_{\text{contra}}}{\partial \mathbf{f}_i} = \sum_{v \in \mathcal{V}(i)} \frac{\partial \mathcal{L}_{\text{contra}}^{(v)}}{\partial \mathbf{f}_i}, \quad (6)$$

where $\mathcal{V}(i)$ denotes the set of views in which Gaussian primitive i is visible. This parameter sharing couples the supervision provided by different views within a single 3D feature field and thereby promotes cross-view feature consistency, without any explicit cross-view instance matching. In this study, cross-view integration is achieved by jointly optimizing a shared 3D representation with per-view supervision, rather than through explicit inter-view matching, object tracking, or positive-pair construction.

To filter noisy labels, we design an uncertainty estimation module. Let K_u denote the instance ID assigned to pixel u by the SAM3 mask, which serves as the pseudo-label. The uncertainty weight is computed as:

$$\omega_u = 1 - \frac{\exp(\text{sim}(\mathbf{F}_u, \bar{\mathbf{F}}_{K_u}))}{\sum_{l=1}^C \exp(\text{sim}(\mathbf{F}_u, \bar{\mathbf{F}}_l))} \quad (7)$$

A pixel is regarded as reliable when its uncertainty weight is below a threshold τ ; that is, the reliable pixel set used for supervision is $\mathcal{P}_{\text{reliable}} = \{u \mid \omega_u \leq \tau\}$, and pixels with $\omega_u > \tau$ are treated as unreliable and excluded from the contrastive and clustering losses. The choice of the threshold $\tau = 0.5$ is critical for maintaining robustness across different crop categories. This is because weak-contrast cases, such as green sweet peppers against green foliage, and strong-contrast cases, such as red apples, exhibit different levels of tolerance to segmentation noise. To examine this effect, we systematically evaluate $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ on six representative crop categories. The results are reported in Table 3.

The sensitivity analysis for τ provides findings similar to those reported in Table 3. The best performance values for plum and mango are obtained at $\tau = 0.3$, indicating that more boundary pixels need to be retained when the foreground-background contrast is weak. In contrast, apple and peach achieve their best results at $\tau = 0.5$. Across all six crop categories, the highest average F1 score is 0.977, which

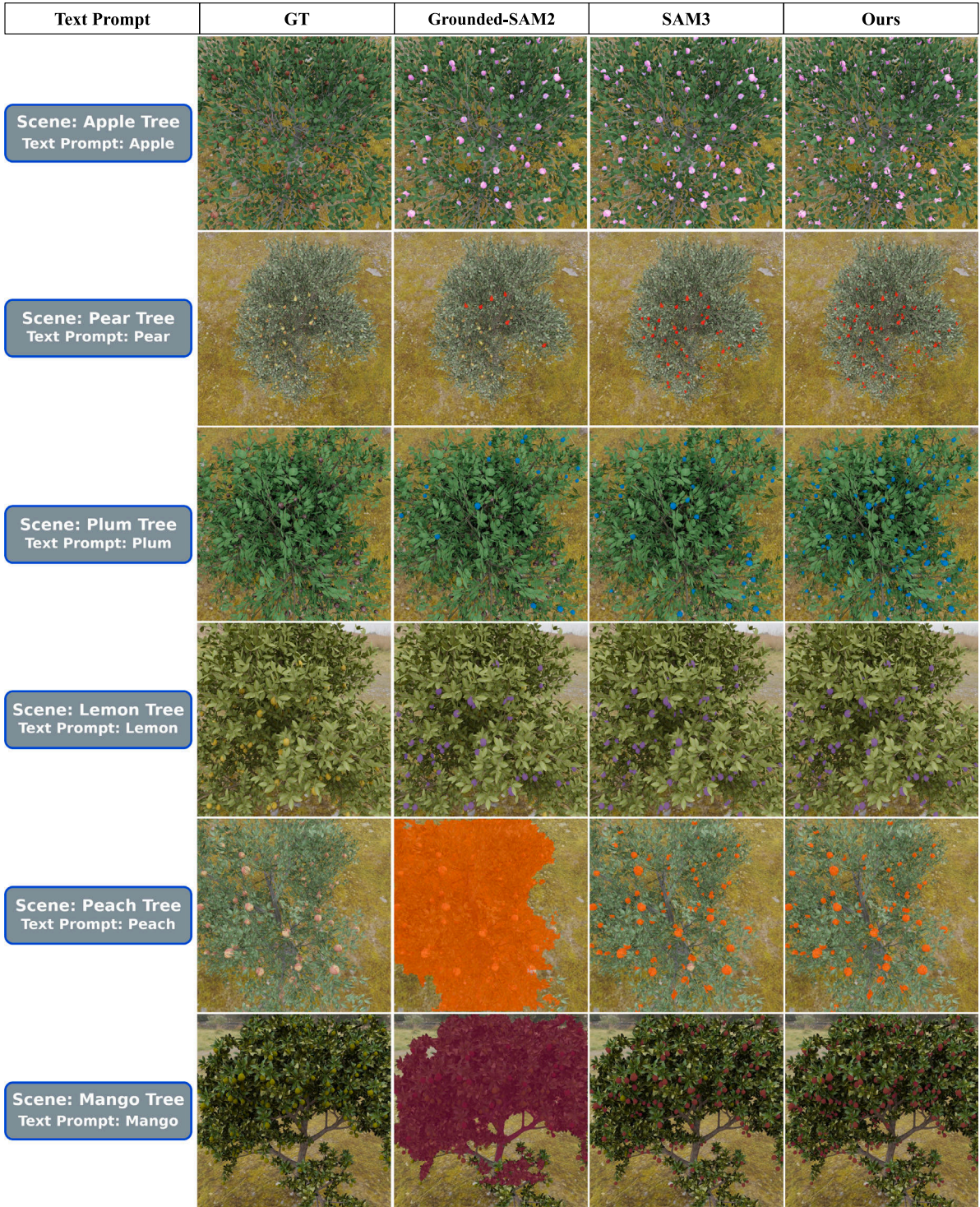


Fig. 5. Qualitative comparison of fruit segmentation on synthetic FruitNeRF scenes, including apple, pear, lemon, plum, peach, and mango. Grounded-SAM2 produces clear false positives in the peach scene and misclassifies almost the entire mango tree canopy as fruit. SAM3 reduces some background errors but still suffers from fragmented and over-segmented predictions. In contrast, our improved SAM3 generates more accurate and coherent fruit masks across different fruit categories.

is also obtained at $\tau = 0.5$. Within the range of $\tau \in [0.4, 0.6]$, the performance variation does not exceed 1.5%, suggesting that $\tau = 0.5$ can be considered a stable parameter setting for different crops. When τ is too high, for instance at $\tau = 0.7$, the F1 score for plum decreases to 0.882, indicating that an overly aggressive threshold may remove useful fruit regions.

2.3.2. Object-level codebook and clustering loss

Per-view instance identities cannot serve directly as global supervision, because a promptable segmenter returns identities that are local to each image; an intermediate scene-level representation is therefore required to bind them together. To obtain object-level instance assignments from the learned Gaussian features, we introduce a learnable

Table 3Sensitivity of the uncertainty threshold τ on the FruitNeRF synthetic dataset. Best value per column in bold.

Threshold τ	Apple F1	Plum F1	Lemon F1	Pear F1	Peach F1	Mango F1	Avg. F1
0.3	0.951	0.968	0.928	0.943	0.932	0.974	0.949
0.4	0.978	0.962	0.958	0.971	0.964	0.969	0.967
0.5 ^a	0.996	0.967	0.971	0.981	0.977	0.971	0.977
0.6	0.984	0.934	0.964	0.973	0.969	0.948	0.962
0.7	0.946	0.882	0.937	0.945	0.938	0.893	0.924

^a Marks the default configuration.

object-level codebook $\mathbf{F}_{\text{obj}} \in \mathbb{R}^{256 \times 16}$, where 256 is the number of codewords and 16 is the feature dimension that matches the dimensionality of the Gaussian feature vectors. Each codeword, i.e., each row of $\mathbf{F}_{\text{obj}}$, represents a candidate fruit instance prototype in the scene. The codebook serves as an intermediate representation during training, which maps the rendered pixel-level Gaussian features to different object prototypes. It is jointly optimized with the Gaussian features through backpropagation and does not rely on crop-specific shape or size priors. Given the rendered feature map, we first flatten the per-pixel features along the spatial dimensions and compute the pixel-codebook response as:

$$\mathbf{S} = \mathbf{F}_{\text{obj}} \cdot \mathbf{F}_{\text{flatten}}, \quad \mathbf{P}_u = \text{softmax}(\mathbf{S}_{:,u}). \quad (8)$$

Here, $\mathbf{F}_{\text{flatten}} \in \mathbb{R}^{16 \times HW}$ denotes the flattened pixel-level feature matrix, where H and W are the height and width of the rendered feature map, and the operator $\cdot$ denotes matrix multiplication rather than an element-wise product, so that $\mathbf{S} \in \mathbb{R}^{256 \times HW}$. Accordingly, $\mathbf{S}_{:,u} \in \mathbb{R}^{256}$ denotes the response vector of pixel u to all 256 codewords, and $\mathbf{P}_u = \text{softmax}(\mathbf{S}_{:,u}) \in \mathbb{R}^{256}$ is the resulting probability distribution of pixel u over the codewords. During inference, the instance ID of pixel u is assigned by selecting the codeword with the highest response, where $l \in \{1, \dots, 256\}$ indexes the codewords and $\mathbf{P}_u[l]$ is the l th entry of $\mathbf{P}_u$:

$$\hat{K}_u = \arg \max_l \mathbf{P}_u[l]. \quad (9)$$

Since the instance IDs predicted by SAM3 are local to each image and are not globally consistent across views, directly aligning these IDs in multi-view images may introduce label drift during training. For example, the same fruit may be assigned different instance IDs in different views. To avoid this problem, we perform the assignment between SAM3 instances and codebook prototypes within each individual view. This assignment is a one-to-one matching between the SAM3 instances and the object-level codewords inside a single frame, and is used only to define the supervision target of the clustering loss; it does not perform any temporal or cross-view tracking of fruit instances. Given the SAM3 instance partition $\{\Omega_j\}_{j=1}^C$ in the current view, we construct the following cost matrix for Hungarian matching:

$$C_{jl} = - \sum_{u \in \Omega_j} \mathbf{P}_u[l]. \quad (10)$$

This cost measures the overall response of pixels belonging to instance Ω_j to the l th codeword. If the pixels within an instance show a stronger response to a certain codeword, assigning this instance to that codeword produces a lower cost. Solving the resulting linear assignment problem gives the optimal mapping:

$$\Pi^* = \arg \min_{\Pi} \sum_{j=1}^C C_{j, \Pi(j)}. \quad (11)$$

Here, Π^* assigns each SAM3 instance in the current view to a unique object prototype. To reduce the influence of noisy pseudo-labels, the clustering loss is computed only on reliable pixels selected by the uncertainty-aware filtering strategy described in Section 2.3.1. This avoids using ambiguous boundary pixels, occluded regions, or background-like responses as strong supervision for object-level prototype learning. Based on the Hungarian assignment, the clustering loss is defined as:

$$\mathcal{L}_{\text{cluster}} = \frac{1}{|\mathcal{P}_{\text{reliable}}|} \sum_{u \in \mathcal{P}_{\text{reliable}}} \left\| \mathbf{F}_{\text{obj}}[\Pi^*(K_u)] - \frac{\mathbf{F}_u}{\|\mathbf{F}_u\|_2} \right\|. \quad (12)$$

Here, K_u denotes the SAM3 instance ID of pixel u , and $\mathcal{P}_{\text{reliable}}$ denotes the reliable pixel set obtained from the uncertainty-aware filtering strategy. This loss encourages the assigned object prototype and the normalized rendered Gaussian feature to have consistent directions in the feature space. As a result, Gaussian primitives belonging to the same fruit instance are pulled closer in the feature space, while different fruit instances become easier to separate.

2.3.3. 3D fruit localization based on hough voting

The SAM3 masks may contain spatial noise caused by leaf occlusion, blurred boundaries, and inconsistent instance labels across different viewpoints, making direct use of the projected Gaussian primitives unreliable. Moreover, in 3DGS, Gaussian primitives mainly represent the visible object surfaces rather than the complete fruit volume. Therefore, directly clustering the original Gaussian positions is highly sensitive to mask noise and boundary errors.

To address this issue, we propose a voting-based method to fuse spatial information from multiple viewpoints. Each Gaussian primitive is assigned a learnable 3D offset $\Delta \mathbf{p}_i$, which shifts its center toward a potential fruit instance center. The corresponding *Gaussian-level voting point* is defined as $\mathbf{V}_i^{3D} = \mathbf{p}_i + \Delta \mathbf{p}_i$. Through this voting process, Gaussian primitives belonging to the same fruit are encouraged to produce voting points that converge to a common instance center.

The offset is parameterized by a learnable variable $\hat{\mathbf{o}}_i$ that is optimized jointly with position, scale, rotation, opacity, color, and the feature vector, rather than being predicted by an additional network head. It is initialized to zero, and its magnitude is bounded:

$$\mathbf{V}_i^{3D} = \mathbf{p}_i + \Delta \mathbf{p}_i, \quad \Delta \mathbf{p}_i = \delta_{\max} \tanh(\hat{\mathbf{o}}_i), \quad \Delta \mathbf{p}_i|_{t=0} = \mathbf{0}, \quad (13)$$

where $\delta_{\max} = 0.05 d_{\text{scene}}$ and d_{scene} is the diagonal length of the bounding box of the COLMAP sparse point cloud. The hyperbolic tangent is applied coordinate-wise, so every offset component is confined to $[-\delta_{\max}, \delta_{\max}]$, which gives $\|\Delta \mathbf{p}_i\|_{\infty} \leq \delta_{\max}$ and $\|\Delta \mathbf{p}_i\|_2 \leq \sqrt{3} \delta_{\max}$. The offset parameters use the same learning rate and decay schedule as the Gaussian positions, and new primitives created by densification inherit the offset of their parent. This constraint bounds the displacement of a voting point relative to the scale of the scene and prevents background primitives from producing large non-physical offsets.

For each pixel u , the visible Gaussian-level voting points are aggregated by averaging:

$$\mathbf{V}_u^{3D} = \frac{1}{|\mathcal{M}_u|} \sum_{i \in \mathcal{M}_u} \mathbf{V}_i^{3D} \quad (14)$$

where $\mathcal{M}_u$ denotes the set of Gaussian primitives that are visible at pixel u , i.e., those that contribute to the rendering of pixel u , and $|\mathcal{M}_u|$ is its cardinality.

Supervision comes from the 2D instance centroids of the SAM3 masks. Let $\mathbf{m}_{K_u}^{2D}$ denote the 2D centroid of the SAM3 mask (indexed by K_u) assigned to pixel u , and let $\pi(\cdot) : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ denote the camera projection operator that maps a 3D point to its 2D image-plane coordinates using the camera intrinsics and extrinsics, including the perspective division. The voting loss is then defined as

$$\mathcal{L}_{\text{vote}} = \frac{1}{|\mathcal{P}|} \sum_{u \in \mathcal{P}} \left\| \pi(\mathbf{V}_u^{3D}) - \mathbf{m}_{K_u}^{2D} \right\| \quad (15)$$

where $\mathcal{P}$ denotes the set of pixels used for voting supervision. This loss indirectly trains the per-Gaussian 3D offset by projecting the aggregated

3D voting point back to 2D and aligning it with the corresponding mask centroid.

After training convergence, Gaussian primitives located on leaves, branches, or background regions may still produce unstable or semantically weak voting responses. Therefore, before instance clustering, we remove unreliable primitives according to two criteria: the cross-view dispersion of their Gaussian-level voting points and their maximum response to the object-level codebook.

Let $\mathcal{V}(i)$ denote the set of views in which primitive i is judged visible by the rasterizer and can be associated with a valid SAM3 fruit mask, and let $\mathbf{m}_i^{(v)}$ be the centroid of the corresponding instance mask in view v . The reprojection residual of the 3D voting point in that view and its mean over the visible views are

$$\mathbf{r}_i^{(v)} = \pi_v(\mathbf{V}_i^{3D}) - \mathbf{m}_i^{(v)}, \quad \bar{\mathbf{r}}_i = \frac{1}{|\mathcal{V}(i)|} \sum_{v \in \mathcal{V}(i)} \mathbf{r}_i^{(v)}, \quad (16)$$

where $\pi_v(\cdot)$ is the camera projection of view v . The cross-view voting dispersion is then

$$\sigma_i = \left[\frac{1}{|\mathcal{V}(i)|} \sum_{v \in \mathcal{V}(i)} \left\| \mathbf{r}_i^{(v)} - \bar{\mathbf{r}}_i \right\|_2^2 \right]^{1/2}, \quad (17)$$

measured in pixels. Primitives that lie on fruit produce consistent reprojection residuals across views and therefore have a small σ_i , whereas the residuals of leaves, branches, and other background primitives vary strongly with viewpoint. Accordingly, primitives with σ_i above a fixed threshold σ_{thr} are labeled as background and discarded before clustering. The threshold is determined once on the synthetic scenes and is then held fixed for all datasets, including the real orchard and greenhouse scenes; it is not re-tuned per scene or per crop, so the final fruit count does not depend on any scene-specific threshold adjustment. Primitives with inconsistent voting offsets or low object-level confidence are treated as background and excluded from subsequent clustering. The remaining Gaussian-level voting points $\{\mathbf{V}_i^{3D}\}$ are then clustered using HDBSCAN (McInnes et al., 2017) to obtain fruit-instance hypotheses.

HDBSCAN is adopted because the density of fruit-related voting points is often non-uniform in orchard and greenhouse scenes. Unlike DBSCAN, which requires a manually specified global density threshold, HDBSCAN builds a hierarchical cluster structure based on mutual reachability distance and selects stable clusters according to the Excess of Mass criterion. This property allows the method to handle fruit instances with different point densities caused by occlusion, viewpoint changes, and varying fruit sizes. As illustrated in Fig. 6, dense voting points are grouped into stable clusters, while isolated or weakly supported points are regarded as noise and removed from counting. In all experiments, we use the same HDBSCAN configuration across FruitNeRF synthetic scenes, FruitNeRF real scenes, Fuji-SfM, and BUP20, without crop-specific adjustment. The retained voting points are aligned to a common metric scale by a Sim(3) transform before clustering, and no anisotropic normalization is applied. We set `min_cluster_size` = 30, `min_samples` = 10, and `alpha` = 1.0. Points assigned the noise label -1 are excluded from counting, the number of valid clusters is taken as the predicted fruit count, and the 3D center of each fruit is the mean of the voting points in its cluster.

Let $\{C_k\}_{k=1}^K$ denote the set of valid clusters returned by HDBSCAN, where points assigned to the noise label -1 are excluded. The fruit count is defined as the number of valid clusters K , and the 3D center of the k th fruit is estimated by computing the centroid of its corresponding voting points:

$$\mathbf{c}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{V}_i^{3D}. \quad (18)$$

In contrast, FruitNeRF (Meyer et al., 2024) adopts a cascaded clustering scheme that combines DBSCAN, agglomerative clustering, and Hausdorff-distance template matching. Its performance relies on several crop-specific geometric priors, including a template fruit point cloud,

the average fruit radius, and an upper bound on the number of fruits per cluster, all of which must be re-specified for a new crop type, limiting the generalizability of the approach. Our clustering pipeline introduces no priors related to fruit shape or size, and can therefore be transferred directly from apples in open orchards to sweet peppers in greenhouses without re-tuning any parameter at the clustering stage; only the SAM3 text prompt is changed, while the 3D Gaussian representation is still optimized for each new scene. This is empirically supported by the BUP20 results reported in Section 3.

Clustering these voting-refined 3D points rather than the direct SAM3 segmentation masks or per-view Gaussian primitives makes the instance segmentation results less sensitive to mask noise, occlusions, and inaccurate boundaries. The full voting pipeline, shown in Fig. 7, integrates 2D mask-guided segmentation, multi-view voting, and background filtering into a single process. It should be noted that voting is not utilized in our work to predict a rigid geometric center of each fruit instance. Instead, the voting process estimates a mask-guided mass center through the supervision of projected centroids of 2D masks. As a result, the learned voting response is more robust for instance grouping rather than accurate geometric center estimation. For fruits with irregular or elongated shapes such as sweet peppers, the visible centroid position can be further affected by occlusion and viewpoint changes. To address this, multi-view voting points are aggregated in 3D and used as clustering cues, which reduces the offset caused by single-view occlusion and shape irregularity. The experiments on the BUP20 sweet pepper dataset have shown that the voting-refined points can be clustered into separate fruit instances.

2.3.4. Joint optimization and end-to-end inference

The complete objective function integrates reconstruction, feature learning, voting, and clustering into a single loss function:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_c \mathcal{L}_{\text{contra}} + \lambda_v \mathcal{L}_{\text{vote}} + \lambda_n \mathcal{L}_{\text{cluster}} \quad (19)$$

where $\mathcal{L}_{\text{recon}} = (1 - \lambda_{\text{dssim}}) \cdot L_1 + \lambda_{\text{dssim}} \cdot (1 - \text{SSIM})$ denotes the photometric reconstruction loss following Kerbl et al. (2023), and λ_c , λ_v , λ_n are the weighting coefficients of the corresponding loss terms. Each loss term plays a different role during training: the reconstruction loss preserves visual consistency between the rendered images and the input views; the contrastive loss improves instance-level feature separation; the voting loss encourages Gaussian primitives to shift toward the fruit center; and the clustering loss associates the learned features with the codebook. By jointly optimizing these losses, information from different modules can interact during training, leading to more stable fruit instance separation and localization. During inference, the feature map F is rendered from the learned Gaussian primitives. The codebook response is then computed as $\mathbf{S} = \mathbf{F}_{\text{obj}} \cdot \mathbf{F}_{\text{flatten}}$, and the instance label for each pixel is assigned as $\hat{K}_u = \arg \max_i \text{softmax}(\mathbf{S}_{:,u})[i]$. This process does not require additional parameter tuning. Finally, the converged Gaussian-level 3D voting points are sequentially processed by the background pruning and HDBSCAN clustering stages described in Section 2.3.3, simultaneously producing the fruit count and the 3D center location of each fruit, which together provide complete counting and spatial information for downstream tasks.

2.4. Experimental settings

2.4.1. Training hyperparameters

The implementation details and training settings are summarized as follows. During the SAM3 LoRA adaptation stage, we use the AdamW optimizer with a learning rate of 1×10^{-4} , a weight decay of 0.01, and a batch size of 4. The model is trained for 30 epochs. The LoRA rank is set to $r = 8$, with a scaling factor of $\alpha = 16$ and a dropout rate of 0.1. For 3DGS optimization, the model is trained for 30,000 iterations. The learning rates for position, feature, and opacity are set to 1.6×10^{-4} , 2.5×10^{-3} , and 0.05, respectively, and the densification interval is set to 100. The loss weights λ_c , λ_v , and λ_n are set to 1×10^{-5} , 1×10^{-6} , and 1×10^{-6} , respectively. All experiments are conducted on a single NVIDIA A6000 GPU with 48 GB memory.

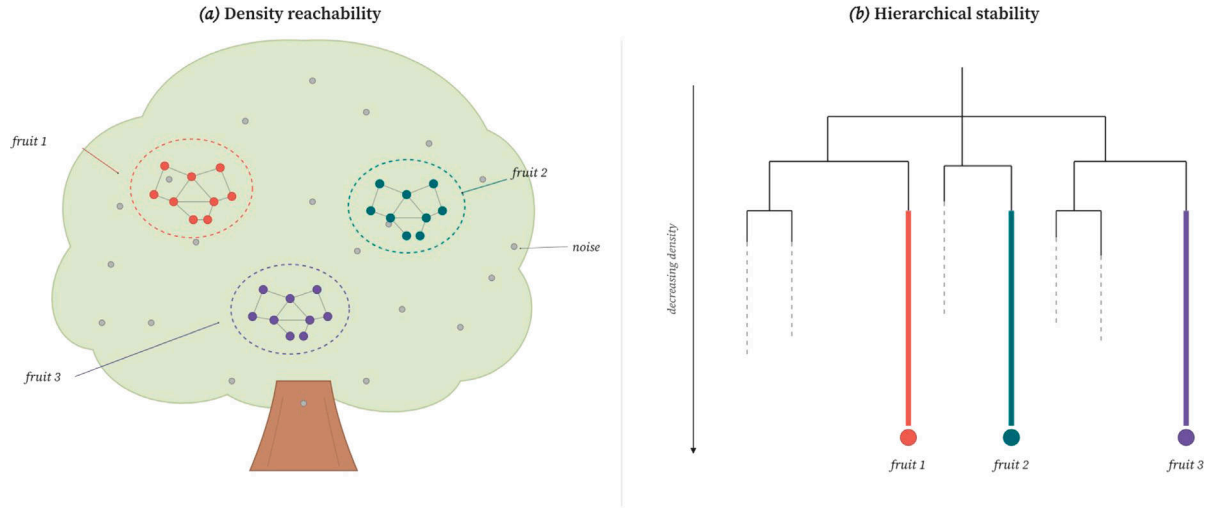


Fig. 6. Conceptual illustration of HDBSCAN-based fruit instance clustering. (a) Voting-refined 3D points are grouped into density-connected clusters, while isolated sparse points are identified as noise. (b) A hierarchical cluster tree is constructed along decreasing density, where stable branches are retained as final fruit instances and unstable candidates are discarded.

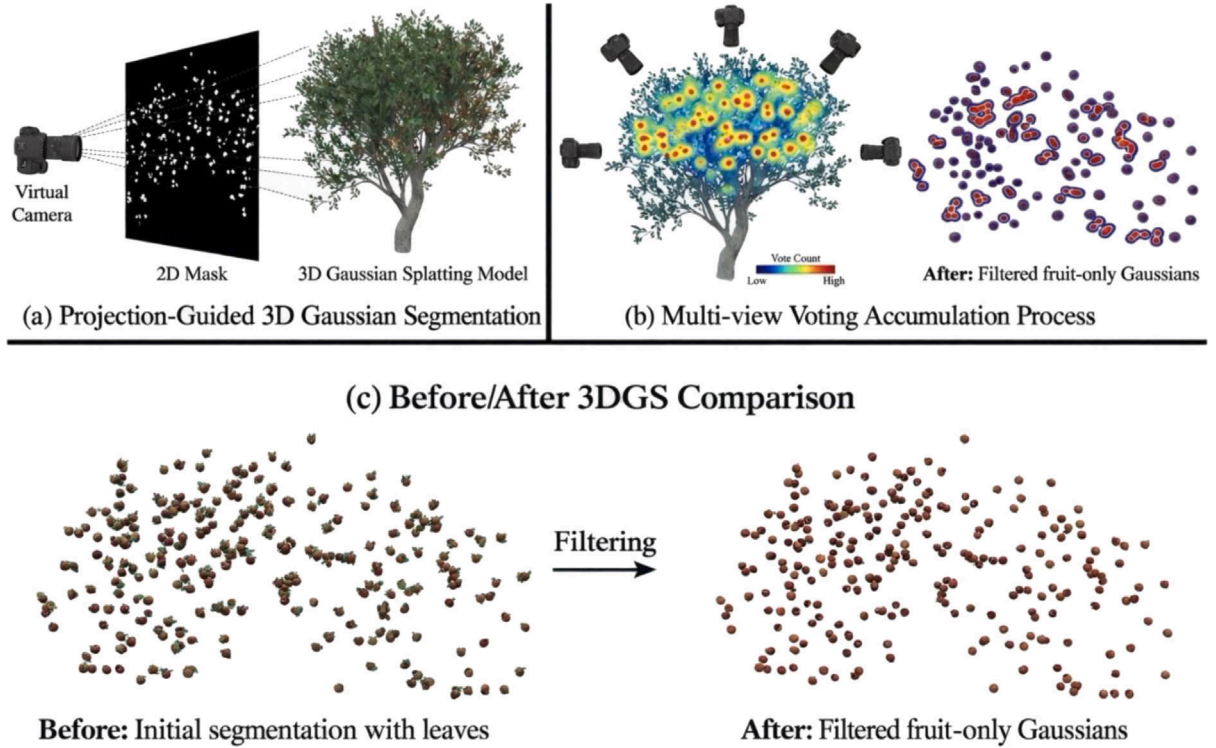


Fig. 7. Hough voting method. (a) 2D mask-guided 3D Gaussian segmentation; (b) Multi-view voting to remove non-fruit Gaussian primitives; (c) Comparison of point clouds before and after filtering.

2.4.2. Evaluation metrics

Counting performance is evaluated using Precision, Recall, F1-score, and mean absolute error (MAE). MAE measures the absolute deviation between the predicted and ground-truth fruit counts, and is defined as

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_{\text{pred}}^i - y_{\text{gt}}^i|, \quad (20)$$

where N is the number of test samples, and y_{pred}^i and y_{gt}^i denote the predicted and ground-truth fruit counts of the i th sample, respectively.

The per-category absolute error (AE) is the individual term of this sum, and the MAE is its mean over the evaluated categories.

For the synthetic scenes, where ground-truth 3D fruit centers are available, we also report the 3D center localization error. Predicted and ground-truth centers are matched one-to-one with the Hungarian algorithm on their Euclidean distances, and a match is valid only if the distance does not exceed the mean fruit diameter of that category. The center error is the mean distance over all valid matches; unmatched predictions and unmatched ground-truth instances are counted as false positives and false negatives, so that the center error, Precision, and Recall all derive from the same matching.

Table 4

Computational efficiency and end-to-end processing time on the FruitNeRF synthetic plum scene (300 frames), NVIDIA RTX A6000. Peak GPU memory is reported; best result in bold.

Method	Backbone	SfM	Training	Post-processing	End-to-end	Rendering FPS ^a	GPU Memory	F1 (Plum)
FruitNeRF (Meyer et al., 2024)	NeRF	35 min	2.5 h	13 min	3.3 h	<0.5	24 GB	0.575
FruitNeRF++ (Meyer et al., 2025)	NeRF + Contrastive	38 min	~8 h	18 min	8.9 h	<0.5	24 GB	0.859
SAM3-Adapt (Ours) ★	3DGS + LoRA SAM3	36 min	1.5 h	15 min	2.4 h	~110	21 GB	0.971

^a Novel-view rendering only.

For 3D instance segmentation evaluation, each predicted fruit instance is represented by a cluster of Gaussian primitives and evaluated against the ground-truth instance annotations using 3D IoU. We perform instance matching based on the IoU between predicted and ground-truth instances. A prediction is considered a true positive (TP) when its IoU with a ground-truth instance exceeds a predefined threshold. Unmatched predictions are counted as false positives (FP), and unmatched ground-truth instances are counted as false negatives (FN). Following this evaluation protocol, we report mIoU, mAcc, mAP, AP@0.25, and AP@0.50. Here, mIoU denotes the mean IoU over matched instances, and mAcc reflects the instance-level matching accuracy. The mAP is computed over IoU thresholds from 0.50 to 0.95 with a step size of 0.05, while AP@0.25 and AP@0.50 are reported at fixed IoU thresholds of 0.25 and 0.50, respectively.

3. Results

3.1. Dataset splits and computational cost

We conduct experiments on three datasets, with the dataset composition and splitting strategy described in Section 2.1.1. All experiments are run on a single NVIDIA RTX A6000 GPU. Camera pose estimation and 3D reconstruction can take a large amount of time in large orchard scenes. Therefore, we also report the running time of each part of the full pipeline. For the Tree03 scene with 300 input frames, COLMAP sparse reconstruction takes about 38 min, 2D mask inference with the adapted SAM3 model takes about 12 min, and 3DGS training for 30k iterations takes around 1.5 h. For the larger Fuji-SfM scene with 1455 fruit instances, COLMAP sparse reconstruction takes nearly 4 h, which is about 45% of the total pipeline time. This shows that camera pose estimation is still a major time-consuming step in large-scale 3D fruit counting.

The SAM3 LoRA adaptation stage takes about 3 h. However, this cost is only needed once and is not tied to a specific test scene. After training, the LoRA weights can be reused directly for different orchard or greenhouse scenes. Therefore, this stage is not counted as repeated per-scene cost when reporting the running time of a single scene. The overall efficiency comparison of different methods is reported in Table 4.

End to end, the proposed method completed the full pipeline in 2.4 h on the 300-frame scene, compared with 3.3 h for FruitNeRF and 8.9 h for FruitNeRF++ (Table 4). Each pipeline comprised three stages: COLMAP-based sparse reconstruction (SfM), scene optimization, and post-processing, including mask inference, point-cloud or feature export, clustering, and counting. COLMAP was run independently for each method using identical input images and reconstruction parameters. Under this common protocol, SfM and post-processing required 35–38 min and 13–18 min, respectively, across the three methods. Rendering throughput was measured only after scene optimization and excluded offline voting and HDBSCAN clustering; it therefore should not be interpreted as end-to-end efficiency. Benefiting from the explicit 3DGS representation, the proposed method achieves approximately 110 FPS, which is about two orders of magnitude faster than NeRF-based methods. This substantial improvement in rendering efficiency is important for future deployment in real-world agricultural scenarios, where fast visualization and repeated scene inspection are often

required. Consistent with the timings above, camera pose estimation remains a substantial fraction of the total cost and becomes the dominant bottleneck in larger scenes such as Fuji-SfM.

3.2. Fruit counting performance

3.2.1. Synthetic dataset results

On the FruitNeRF synthetic dataset, our method performs better than FruitNeRF in all six fruit categories, and the improvement is more obvious in dense scenes, as shown in Table 5. In the apple scene, our method detects all 283 fruits and obtains an F1-score of 1.000. The difference becomes larger when fruits are highly clustered. In the plum scene, there are 781 fruit instances, and the distance between nearby fruits is smaller than 5 cm. FruitNeRF detects only 315 fruits with an F1-score of 0.575, mainly because overlapping projections tend to merge several fruits into one detection. By comparison, our method detects 741 fruits and reaches an F1-score of 0.971, giving a relative improvement of 68.9%. A similar result can be found in the mango scene, which contains 1150 heavily occluded instances. FruitNeRF detects 807 fruits with an F1-score of 0.816, while our method detects 1123 fruits and obtains an F1-score of 0.977.

Table 5 additionally reports the absolute counting error and the 3D center localization error; for FruitNeRF, both are computed from its released predictions under the protocol of Section 2.4.2, so the two column groups are directly comparable. Averaged over the six categories, the mean absolute counting error is 13.67 fruits for the proposed method and 139.17 fruits for FruitNeRF. The 3D center localization error ranges from 0.21 to 0.56 cm for the proposed method and from 0.34 to 0.88 cm for FruitNeRF, and the proposed method is closer to the ground-truth center in every category, with the largest margin in the heavily occluded plum and mango scenes. Precision, Recall, and F1 are reported alongside the absolute error because a correct count does not imply that every instance is matched correctly: in the lemon scene the count predicted by FruitNeRF equals the ground truth exactly, yet about six instances remain incorrectly matched.

3.2.2. Real-world dataset results

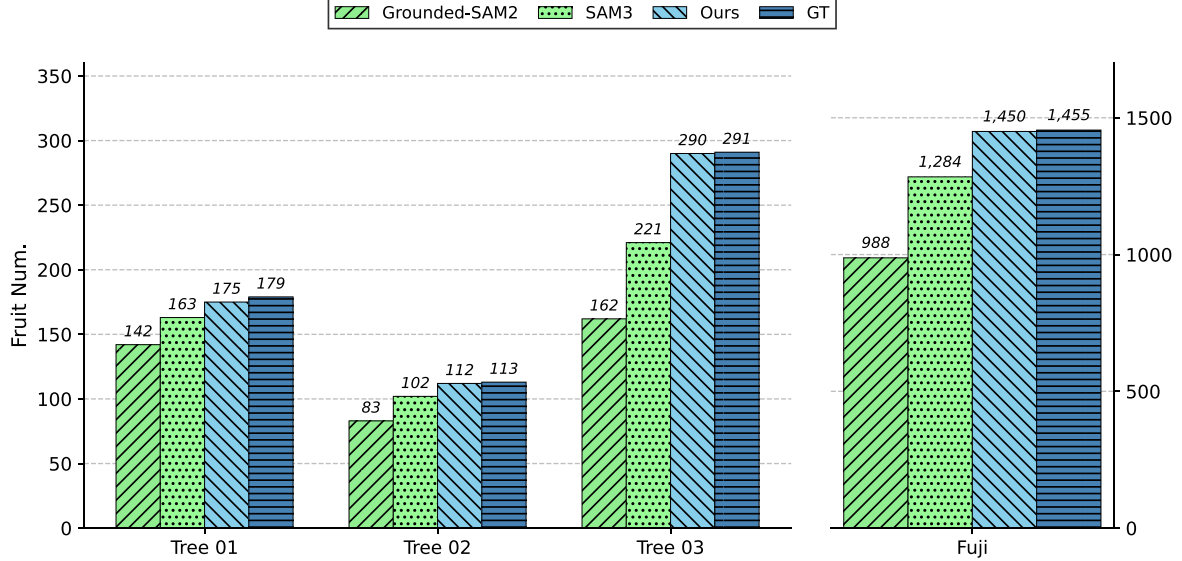
On the four real-world scenes, SAM3-Adapt gives counting results close to the ground truth, with errors within four fruits. It also shows clear advantages in the densely occluded Tree03 scene and the BUP20 scene where the annotations are less precise, as reported in Table 6 and Fig. 8. For Tree01, our method detects 175 fruits, compared with 173 from FruitNeRF and 179 ground-truth fruits. For Tree03, our method detects 290 fruits, very close to the ground truth of 291, whereas FruitNeRF detects only 264 fruits. In the large-scale Fuji-SfM scene with 1455 fruit instances, our method detects 1450 fruits, which is close to FruitNeRF with 1459 detections. In the BUP20 greenhouse scene, our method detects 107 sweet peppers, while FruitNeRF detects only 83. As shown in Fig. 8, we further compare our method with the Grounded-SAM2 and SAM3 baselines, whose 2D masks are fed into the same 3DGS-based voting and clustering pipeline to obtain their 3D counts; our method produces counts consistently closer to the ground truth across all scenes.

Table 5

Fruit counting and 3D center localization on the FruitNeRF synthetic dataset (300 frames per category at 1024×1024). The count closest to the ground truth in each row is highlighted.

Fruit type	FruitNeRF (SAM)						Ours (Adapted SAM3)					
	Count ^a	Precision ^a	Recall ^a	F1-score ^a	Absolute error ^a	Center error (cm) ^a	Count	Precision	Recall	F1-score	Absolute error	Center error (cm)
Apple	282/283	0.992	0.989	0.991	1	0.34	283/283	1.0	1.0	1.0	0	0.29
Plum	315/781	1.0	0.403	0.575	466	0.70	741/781	0.995	0.949	0.971	40	0.21
Lemon	326/326	0.982	0.982	0.982	0	0.41	323/326	0.991	0.957	0.974	3	0.27
Pear	229/250	1.0	0.916	0.956	21	0.52	243/250	1.0	0.972	0.986	7	0.36
Peach	148/152	1.0	0.973	0.987	4	0.38	147/152	0.993	0.967	0.980	5	0.30
Mango	807/1150	0.989	0.694	0.816	343	0.88	1123/1150	0.977	0.976	0.977	27	0.56

^a FruitNeRF counting results from Meyer et al. (2024).

**Fig. 8.** Comparison of 3D fruit counting results in real-world scenes.**Table 6**

3D fruit counting on real-world scenes. The count closest to the ground truth in each row is highlighted.

Dataset	FruitNeRF	Ours	Ground truth
Tree01	173	175	179
Tree02	112	112	113
Tree03	264	290	291
Fuji-SfM	1459	1450	1455
BUP20	83	107	–

3.2.3. Ablation study

To separately examine the independent contributions of the four core components, namely LoRA adaptation, cross-view contrastive learning, clustering loss, and Hough voting, we conduct a systematic ablation study on the two most challenging scenes, plum and mango, from the FruitNeRF synthetic dataset. The baseline configuration already uses LoRA-adapted SAM3 together with 3DGS and DBSCAN, in order to isolate the contribution of the 3D components. Other configurations progressively introduce additional components based on this baseline. The results are reported in Table 7.

The ablation study confirms that each introduced component contributes positively to the final performance, as reported in Table 7. Specifically, contrastive learning, clustering loss, and Hough voting improve the average F1-score by +0.063, +0.033, and +0.051, respectively. Hough voting shows a particularly notable effect in the densely occluded plum scene, where the F1-score increases by +0.077 from configuration (C) to configuration (D). This result highlights the importance of explicit 3D center voting for reducing the ambiguity caused by

2D projection overlap. When all three components are combined, the full method improves the average F1-score by +0.147 compared with the baseline.

3.2.4. 3D instance segmentation performance

To further evaluate the 3D reasoning ability of the proposed method, we report explicit 3D instance-level evaluation results in this section. The evaluation protocol is described in Section 2.4.2. Specifically, we compute the 3D IoU between each predicted 3D fruit instance and the corresponding ground-truth instance. TP, FP, and FN are determined under two IoU thresholds, 0.25 and 0.50, and five metrics are then reported, including mIoU, mAcc, mAP, AP@0.25, and AP@0.50. Since real-world scenes lack accurate 3D instance-level ground truth, this evaluation is conducted on the most challenging plum scene from the FruitNeRF synthetic dataset. The results are shown in Table 8.

3.3. 3D reconstruction results

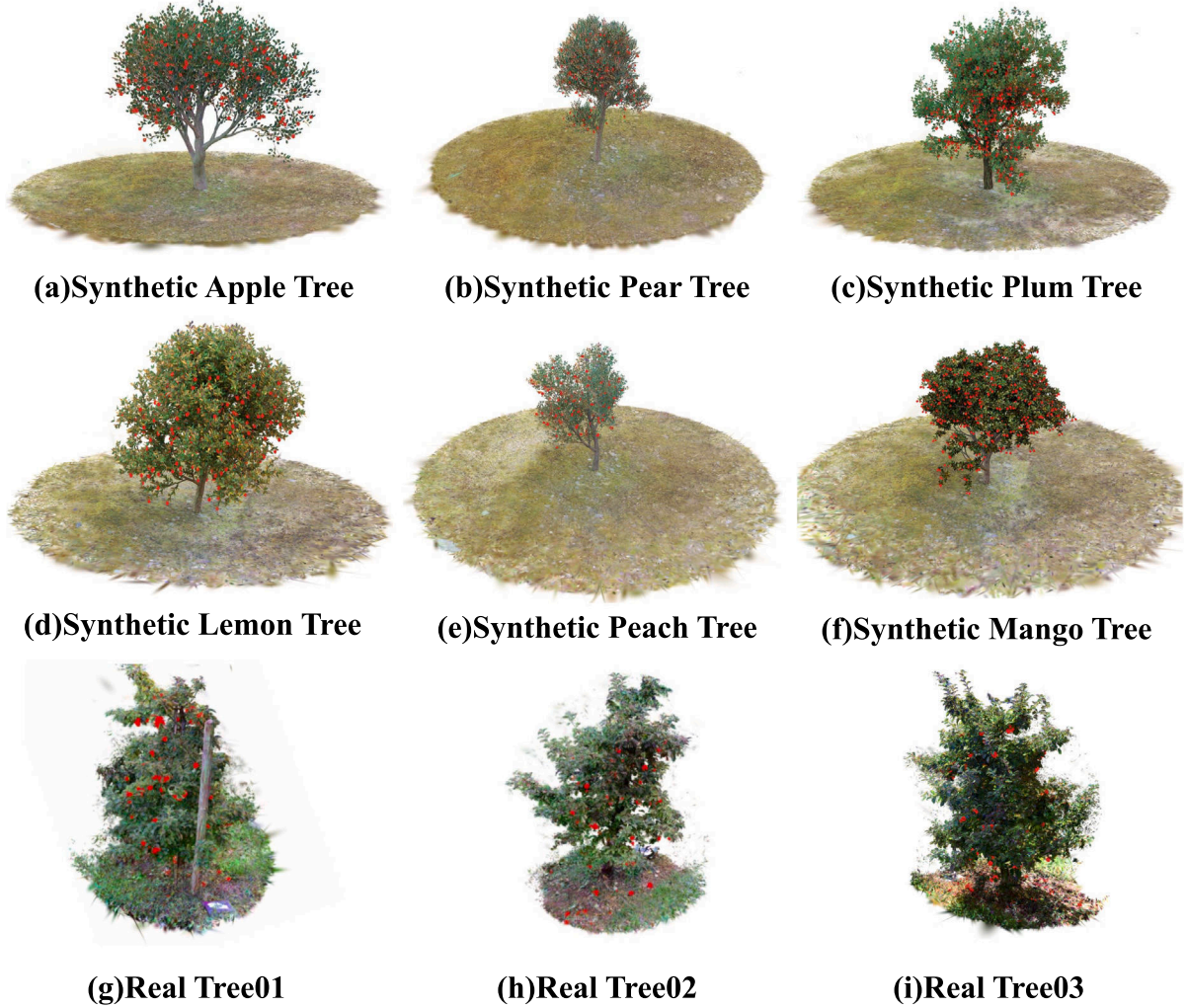
The data sets used in our experiments include diverse settings in terms of data acquisition, varying from synthetic data generated in Blender to real-life orchards with natural lighting changes, foliage, and occlusions. This variety ensures that we have a multi-level environment for evaluation of our method, as illustrated in Fig. 9. While synthetic data sets provide high-quality geometric data due to accurate rendering in Blender, the real FruitNeRF data sets incorporate more realistic factors like different lighting conditions and heavy occlusion.

To further validate our method, we conducted evaluations on two large-scale scenes, which differ significantly from each other in their domains, i.e., Fuji-SfM and BUP20. As seen in Fig. 10, in both cases,

Table 7

Component ablation on the FruitNeRF synthetic plum and mango scenes. Hough voting contributes the most under dense occlusion.

Configuration	LoRA	Contrastive	Clustering	Voting	Plum F1	Mango F1	Avg F1
(A) Baseline: SAM3 + 3DGS + DBSCAN	✓	×	×	×	0.781	0.873	0.827
(B) + Contrastive learning w/ uncertainty weighting	✓	✓	×	×	0.852	0.928	0.890
(C) + Clustering loss	✓	✓	✓	×	0.894	0.951	0.923
(D) Full method (+ Hough voting) ★	✓	✓	✓	✓	0.971	0.977	0.974
Δ from (A)					+0.190	+0.104	+0.147

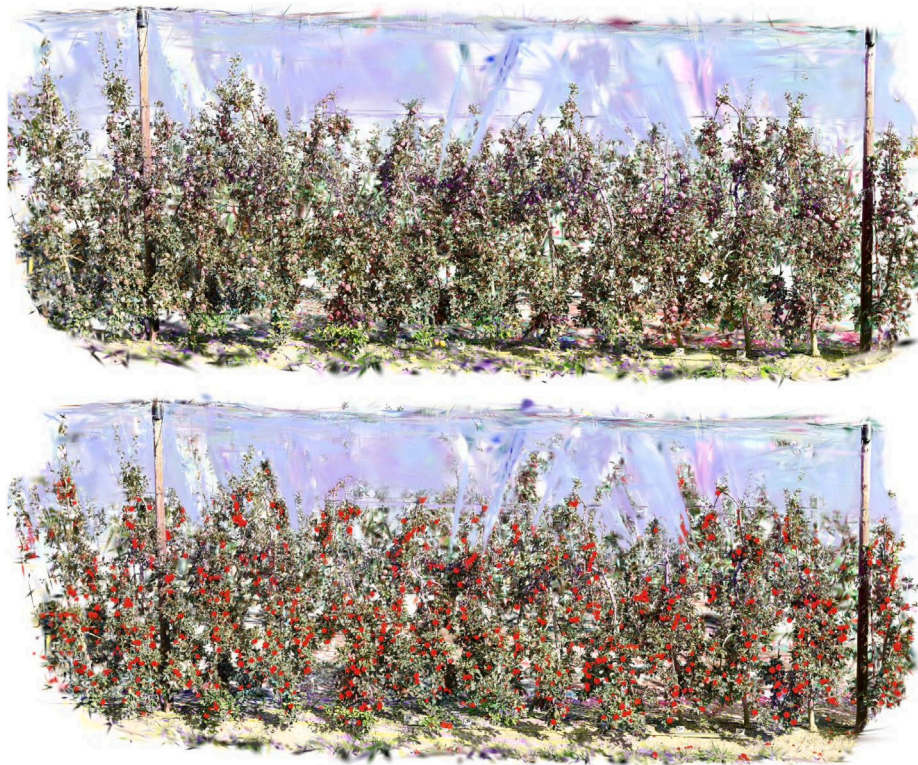
**Fig. 9.** Dataset visualization. (a–c) Apple, pear, and plum scenes from the FruitNeRF synthetic dataset; (d–f) Lemon, peach, and mango scenes; (g–i) Tree01, Tree02, and Tree03 scenes from the FruitNeRF real-world dataset.**Table 8**

3D instance segmentation on the FruitNeRF synthetic plum scene, computed after 3D IoU matching at thresholds of 0.25 and 0.50.

Method	mIoU $\uparrow$	mAcc $\uparrow$	mAP $\uparrow$	AP@0.25 $\uparrow$	AP@0.50 $\uparrow$
Grounded-SAM2	0.382	0.461	0.218	0.337	0.142
FruitNeRF (Meyer et al., 2024)	0.498	0.572	0.341	0.486	0.273
SAM3-Adapt (Ours) ★	0.741	0.823	0.581	0.792	0.562

our method generated highly consistent 3D reconstructions of fruits in the scene. Specifically, Fuji-SfM consists of 1455 apples distributed in an outdoor orchard scene at various depths, while the BUP20 scene

includes greenhouse sweet peppers with very different shapes, textures, and positioning from each other. Nevertheless, the same pipeline is able to generate reasonable results for both scenes.



(a) Fuji-SfM



(b) BUP20 _Data

Fig. 10. 3D reconstruction scenes from Fuji-SfM and BUP20 datasets. (a) Bird's-eye view of large-scale Fuji-SfM orchard scene containing 1455 fruits; (b) BUP20 sweet pepper greenhouse scene, demonstrating cross-crop generalization capability.

4. Discussion

4.1. Counting performance and scene complexity

Experiments on synthetic and real datasets show that the effectiveness of this method is largely determined by the density and occlusion of fruits in the scene. The best improvement is achieved when fruits are very close together, with their boundaries almost completely overlapping on two 2D images. For example, in the plum scene, the F1 score improved from 0.575 to 0.971. This improvement is mainly related to the difference between counting from 2D projections and voting for centers in 3D space. Methods such as FruitNeRF (Meyer et al., 2024) mainly count fruits from rendered 2D masks. When fruits are dense or strongly occluded, several nearby fruits may appear as one connected region in the 2D projection, which easily leads to undercounting. By contrast, our method introduces Hough voting into the 3DGS representation and uses the 3D fruit centers for counting. In this way, fruits that overlap in the image can still be separated in 3D space. As a result, the method does not fully depend on complete 2D object boundaries, but distinguishes instances through their 3D center responses. This change from 2D mask aggregation to 3D center voting explains the clear improvement in dense and complex scenes.

Compared with recent studies on 3D agricultural perception (Jiang et al., 2025; J. Li et al., 2025; Meyer et al., 2025), the proposed method combines foundation model adaptation, cross-view contrastive learning, and 3D voting in one framework. It keeps the real-time rendering advantage of 3DGS, with about 110 FPS, while adding instance segmentation and 3D counting ability. Existing methods are often limited by slower inference, weaker domain adaptation, or insufficient handling of fruit counting under dense occlusion. Compared with recent high-performance 2D detection methods (Z. Liu et al., 2024; Tu et al., 2025; Song et al., 2025; Zhang et al., 2025; Liu et al., 2025), our method uses a 3D representation as an intermediate counting layer. Therefore, it can estimate the fruit number and also provide 3D center coordinates for robotic harvesting and spatial analysis.

4.2. Contribution of each loss term

The ablation study gives a closer look at how the three losses work during training. Fig. 11 plots the fruit-count curves under three training settings on four real-world scenes, namely Tree01, Tree02, Tree03, and Fuji-SfM. When only the contrastive loss is used, the count changes slowly and is often higher than the ground truth. In most cases, it becomes stable only after 25,000 iterations. This shows that cross-view feature consistency is helpful for matching the same fruit across views, but it does not give a clear constraint on the 3D position of fruit centers. Adding the clustering loss makes the training process more stable. The predicted count usually settles between 15,000 and 20,000 iterations. This is because features from the same fruit are pulled closer, while features from different fruits are pushed apart. Still, when fruits are very close or heavily occluded, nearby instances may not be fully separated, which can cause small counting errors. After the voting loss guided by 2D mask centroids is included, the count becomes stable earlier, usually within 10,000–15,000 iterations. The final counting error is within 2%. This trend can also be seen from the green curves in Fig. 11, where the predicted counts move toward the gray dashed ground-truth lines more quickly. This indicates that the voting loss provides a clearer cue for learning 3D fruit centers, especially in dense scenes. More quantitative results are listed in Table 7.

4.3. Transferability across crop types

To further evaluate the transferability of the proposed framework across different crop types and cultivation environments, we apply it to the greenhouse sweet pepper dataset BUP20, in addition to the

two orchard apple datasets. Compared with orchard apples, sweet peppers exhibit substantial differences in plant morphology, fruit appearance, spatial distribution, and imaging conditions. Apples are typically grown in open-field orchard environments, whereas sweet peppers are cultivated under more controlled greenhouse conditions. Therefore, BUP20 provides a representative test case for assessing the cross-crop generalizability of the proposed framework.

The experimental results show that the adapted weights and the post-processing settings transfer from orchard apples to greenhouse sweet peppers: only the SAM3 text prompt is changed, while the LoRA weights and the clustering configuration are kept unchanged, and the 3D Gaussian representation is still optimized for every new scene. This indicates that the method is not limited to orchard apple datasets, but can be transferred to crop types with different visual characteristics and growing environments. From the perspective of crop structure, orchard apples and greenhouse sweet peppers still share certain common properties: in both cases, fruits are generally located within observable canopy regions, and sufficient spatial information can be obtained from multi-view images. These shared properties provide necessary conditions for cross-view feature learning and 3D voting-based instance localization, which partly explains why the framework can be transferred from orchard apple scenes to greenhouse sweet pepper scenes.

However, the applicability of the proposed method to other crop systems still requires further investigation. In particular, vine crops such as grapes and kiwifruits usually have more complex spatial structures, with fruits densely distributed in clusters and very small distances between adjacent instances. In addition, intertwined branches, leaves, and fruit occlusions may further increase the difficulty of instance separation, center localization, and cross-view consistency modeling. Moreover, trellis structures and constrained acquisition paths may limit the available viewpoints, thereby reducing the stability of 3D reconstruction and fruit localization.

It should be noted that the main components of the proposed framework, including LoRA-based vision foundation model adaptation, cross-view contrastive learning, and 3D Hough voting, do not rely on fixed geometric templates or size priors for a specific crop. Therefore, the method has the potential to be extended to other crop types. For vine crops, future work may further incorporate local geometric constraints, cluster-level instance modeling, or separation strategies designed for densely distributed fruits, in order to improve robustness under complex canopy structures and severe occlusions. Recent studies on grape and kiwifruit detection and yield prediction (Devanna et al., 2025; Wang et al., 2025) also provide useful references for extending the proposed framework to vine crop scenarios.

4.4. Limitations

Quantitative evaluation of 3D localization was restricted to the synthetic FruitNeRF scenes because comparable annotations are unavailable for real-world datasets. Neither Fuji-SfM nor BUP20 provides reliable 3D fruit centers or instance-level labels; evaluation on these datasets therefore focused primarily on counting accuracy. Real-world benchmarks containing 3D fruit centers and instance-level annotations are needed to assess localization and segmentation under field conditions. Computational scalability also remains a substantial constraint for orchard-scale deployment. Processing a medium-sized tree required several hours, while COLMAP pose estimation alone required approximately 4 h for the large-scale Fuji-SfM scene. Moreover, each scene requires a separately optimized 3DGS model, limiting scalability to commercial orchards containing thousands of trees. The framework also inherits the static-scene assumption of 3DGS, which may not hold under field conditions. Wind-induced motion, illumination changes, and specular reflections from irrigation or dew may reduce geometric and photometric consistency across views.

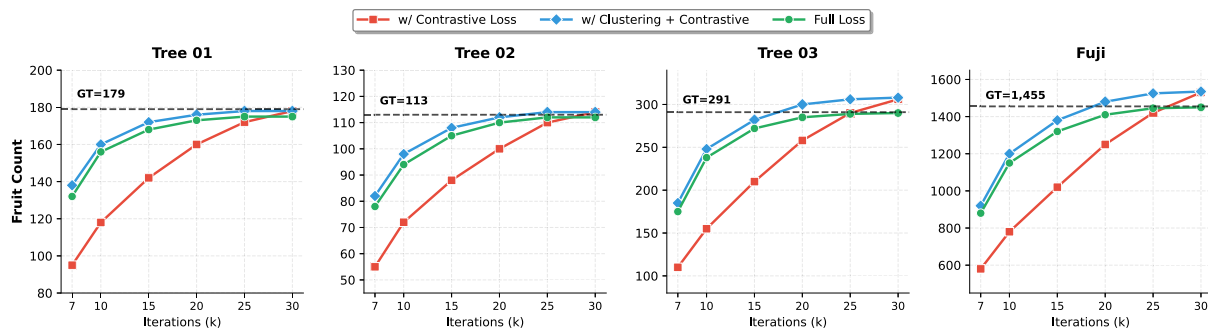


Fig. 11. Convergence curves for fruit counting during training. Red: contrastive loss only; blue: contrastive + clustering loss; green: full loss with voting. Clustering loss accelerates convergence, but accurate final counts require the voting mechanism.

More efficient camera-pose estimation may reduce the dominant computational overhead in large orchard scenes. Learning-based SLAM and global SfM methods, including DROID-SLAM (Teed and Deng, 2021), MAST3r-SfM (Duisterhof et al., 2025), and GLOMAP (Pan et al., 2024), offer potential alternatives to COLMAP. Integrating these approaches with online 3DGS optimization warrants investigation as a route toward deployment on agricultural robotic platforms. Evaluation on vine crops such as grapes and kiwifruit would provide a more stringent test of the framework under dense clustering and severe occlusion. Such scenes complicate fruit-center definition, instance separation, and occlusion modeling across multiple views. Finally, cross-scene pretraining and more generalizable 3DGS representations may reduce per-scene optimization and improve transfer across scenes and crop types.

5. Conclusions

This study addresses fruit counting and 3D localization in orchards and greenhouses, where dense clustering, occlusion, and variable illumination limit the reliability of single-view approaches. We propose a unified framework that combines domain-adapted SAM3 with an explicit 3D Gaussian representation. Low-rank adaptation improves instance segmentation in agricultural imagery while updating only a small subset of model parameters. Uncertainty-weighted cross-view contrastive learning then integrates per-view instance evidence into a shared 3D Gaussian feature space. Hough voting aggregates multi-view geometric cues, followed by clustering to estimate fruit counts and three-dimensional locations. By performing instance discrimination and counting in 3D, the framework exploits complementary views and reduces instance merging, missed detections, and duplicate counts caused by projection overlap.

The results demonstrate that integrating an adapted vision foundation model with an explicit 3D scene representation enables reliable fruit perception across synthetic, orchard, and greenhouse environments. The benefits are particularly evident in densely clustered and heavily occluded scenes. Because clustering and post-processing require no crop-specific shape or size priors, the adapted weights and settings transfer from open-field apples to greenhouse sweet peppers. This transfer requires only a change to the SAM3 text prompt, although the 3D Gaussian representation must be re-optimized for each new scene. The resulting fruit counts and 3D locations provide valuable spatial information for yield estimation and robotic harvesting, with potential applications in other plant phenotyping tasks.

CRedit authorship contribution statement

Kai Zhao: Writing – original draft, Visualization, Validation, Supervision, Methodology, Data curation, Conceptualization. **Chenchen Kang:** Writing – review & editing, Investigation, Funding acquisition, Formal analysis. **Suzy Rogiers:** Resources, Project administration, Investigation, Funding acquisition. **Oula Ghannoum:** Writing – review

& editing, Writing – original draft, Validation, Supervision, Project administration, Funding acquisition. **Yi Guo:** Writing – review & editing, Methodology, Investigation, Funding acquisition.

Ethics declaration

The authors declare that this study did not involve human participants or animal subjects.

Declaration of AI-assisted Writing

During the preparation of this work, the authors used ChatGPT for language polishing and improving clarity. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Funding

Financial support was provided by Future Food Systems Cooperative Research Centre, Australia.

Declaration of competing interest

The authors declare no conflict of interest.

Data availability

The source code, configuration files, and dataset access links are publicly available at <https://github.com/kkk0216/SAM3-3DFruit>.

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